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JEE MAIN PHYSICS DPP: PULLEY SYSTEMS & CONNECTED MASSES

Topic: Advanced Pulley Constraints, Virtual Work Method & Multi-Block Connected Systems | Time: 45 Min

| Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** In a system with a movable pulley P_1 carrying mass $m_1 = 4$ kg and a fixed pulley P_2 , a single light string connects mass $m_2 = 1$ kg on a smooth table to m_1 via P_1 . If the table block moves with acceleration a_2 , the acceleration a_1 of the hanging movable pulley block is:
- $a_1 = 2a_2$
 - $a_1 = \frac{a_2}{2}$
 - $a_1 = a_2$
 - $a_1 = \frac{a_2}{4}$
- Q2.** In the system of Q1, the acceleration a_1 of the hanging mass m_1 is: (Take $g = 10$ m/s²)
- 2.0 m/s²
 - 8.0 m/s²
 - 5.0 m/s²
 - 4.0 m/s²
- Q3.** In an Atwood machine with masses $m_1 = 6$ kg and $m_2 = 4$ kg, the acceleration of the center of mass of the two-mass system (a_{cm}) is: (Take $g = 10$ m/s²)
- 2.0 m/s² downward
 - 0.8 m/s² upward
 - 0.2 m/s² downward
 - 0.4 m/s² downward
- Q4.** A block of mass m_1 on a horizontal table is pulled by a string passing over a fixed pulley at height h . When the string makes an angle θ with the horizontal and the free end is pulled with constant speed u , the speed of block m_1 along the table is:
- $\frac{u}{\cos \theta}$
 - $u \cos \theta$
 - $\frac{u}{\sin \theta}$
 - $u \tan \theta$
- Q5.** In a modified Atwood machine, a hanging mass $m_1 = 2$ kg pulls two masses $m_2 = 3$ kg and $m_3 = 5$ kg tied together by a string on a frictionless table. The tension in the string between m_2 and m_3 is: (Take $g = 10$ m/s²)
- 6 N
 - 10 N
 - 16 N
 - 4 N
- Q6.** A system consists of three blocks A, B , and C of masses $m_A = 1$ kg, $m_B = 2$ kg, and $m_C = 3$ kg connected vertically by strings in a single line. If pulled upward with an acceleration $a = 2$ m/s², the tension in the string connecting A and B (where A is the top block) is: (Take $g = 10$ m/s²)
- 72 N
 - 36 N
 - 60 N
 - 24 N
- Q7.** A block A of mass M slides on a frictionless horizontal floor. A light string attached to A passes over a smooth pulley and holds a hanging block B of mass m . The force exerted by the pulley clamp on the table corner is:
- T
 - $2T$
 - $\frac{T}{\sqrt{2}}$
 - $\sqrt{2}T$
- Q8.** Two masses $m_1 = 3$ kg and $m_2 = 1$ kg are connected by a light string over a smooth pulley. Mass m_1 is placed on a smooth incline of angle $\theta = 30^\circ$, while m_2 hangs vertically. The acceleration of mass m_1 down the incline is: (Take $g = 10$ m/s²)
- 1.25 m/s²
 - 2.50 m/s²
 - 0.75 m/s²
 - 3.75 m/s²
- Q9.** In a virtual work constraint problem, if $\sum \vec{T}_i \cdot \vec{a}_i = 0$ is applied to a connected pulley system, the underlying physical assumption is that:
- Pulleys are frictionless but have finite rotational inertia
 - Strings are massless, inextensible, and pulleys are ideal
 - The acceleration of all masses is identical
 - Mechanical energy is dissipated as heat
- Q10.** A block of mass m is suspended by a string wrapped around two identical fixed pulleys and one movable pulley. If the free end of the string is pulled down with acceleration a_0 , the upward acceleration of mass m is:

- (a) a_0
 (b) $2a_0$
 (c) $\frac{a_0}{2}$
 (d) $\frac{a_0}{4}$
- Q11.** A monkey of mass $m = 30$ kg climbs up a light rope attached to a counterweight block of mass $M = 30$ kg over a smooth fixed pulley. If the monkey accelerates up the rope at 2 m/s^2 relative to the rope, the acceleration of the block M relative to the ground is:
- (a) 2.0 m/s^2 downward
 (b) 0 m/s^2
 (c) 2.0 m/s^2 upward
 (d) 1.0 m/s^2 upward
- Q12.** Two blocks of masses $m_1 = 2$ kg and $m_2 = 3$ kg are connected by a spring of stiffness $k = 100 \text{ N/m}$ on a frictionless horizontal plane. If a constant force $F = 20 \text{ N}$ pulls m_2 horizontally, the maximum extension of the spring from its natural length is:
- (a) 0.16 m
 (b) 0.08 m
 (c) 0.20 m
 (d) 0.32 m
- Q13.** A double pulley system has a fixed pulley and a movable pulley. Mass $m_1 = 4$ kg hangs from the movable pulley, while mass $m_2 = 1$ kg hangs from the free end of the string. The tension in the string is: (Take $g = 10 \text{ m/s}^2$)
- (a) 20.0 N
 (b) 16.0 N
 (c) 24.0 N
 (d) 12.5 N
- Q14.** In Q13, the downward acceleration of the hanging mass m_1 attached to the movable pulley is:
- (a) 4.0 m/s^2
 (b) 1.0 m/s^2
 (c) 2.0 m/s^2
 (d) 5.0 m/s^2
- Q15.** A block of mass $m_1 = 5$ kg is on a smooth horizontal table. It is connected by a string passing over a smooth pulley to a movable pulley of mass $m_p = 0$. The movable pulley supports two hanging masses $m_2 = 2$ kg and $m_3 = 3$ kg. The tension in the string supporting m_1 is: (Take $g = 10 \text{ m/s}^2$)
- (a) 24.0 N
 (b) 36.0 N
 (c) 30.0 N
 (d) 28.2 N
- Q16.** A block of mass $M = 10$ kg is resting on a frictionless horizontal plane. A light string attached to M passes over a frictionless pulley and carries a block of mass $m = 2$ kg. If the system is released from rest, the distance traveled by block M in $t = 2$ seconds is: (Take $g = 10 \text{ m/s}^2$)
- (a) 3.33 m
 (b) 1.67 m
 (c) 6.67 m
 (d) 5.00 m
- Q17.** In a system of connected masses on a smooth wedge of inclination $\theta = 45^\circ$, mass $m_1 = 4$ kg on the incline is connected via a light string over the apex pulley to a hanging mass $m_2 = 4$ kg. The tension in the connecting string is: (Take $g = 10 \text{ m/s}^2$, $\sqrt{2} \approx 1.414$)
- (a) 20.0 N
 (b) 34.1 N
 (c) 40.0 N
 (d) 28.3 N
- Q18.** Three masses $m_1 = 2$ kg, $m_2 = 3$ kg, and $m_3 = 5$ kg are connected by strings as an Atwood machine where m_1 and m_2 hang together on one side and m_3 hangs on the other. The tension in the string connecting m_1 and m_2 is: (Take $g = 10 \text{ m/s}^2$)
- (a) 50 N
 (b) 30 N
 (c) 20 N
 (d) 10 N
- Q19.** A ring of mass m slides along a smooth fixed vertical rod. It is connected by a string passing over a smooth pulley at distance d from the rod to a hanging mass M . When the string makes an angle θ with the vertical, the ring is in static equilibrium if:
- (a) $\sin \theta = \frac{m}{M}$
 (b) $\tan \theta = \frac{m}{M}$
 (c) $\sec \theta = \frac{m}{M}$
 (d) $\cos \theta = \frac{m}{M}$
- Q20.** Two particles of masses m and $2m$ are connected by a light string passing over a frictionless pulley. If the system is initially held at rest and then released, the velocity of the center of mass after time t is:
- (a) $\frac{gt}{9}$ downward
 (b) $\frac{gt}{3}$ downward
 (c) $\frac{2gt}{3}$ downward
 (d) $\frac{gt}{6}$ downward

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(c)	11	(d)	16	(a)
2	(c)	7	(d)	12	(a)	17	(b)
3	(d)	8	(a)	13	(b)	18	(c)
4	(a)	9	(b)	14	(c)	19	(d)
5	(b)	10	(c)	15	(d)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

Length of string $l = x_2 + 2y_1 + \text{constant}$.
Differentiating twice with respect to time gives the kinematic constraint:

$$a_2 + 2a_1 = 0 \implies a_1 = \frac{a_2}{2}$$

Sol 2. Correct Option: (c)

Let T be the tension in the string. For block m_2 on the table: $T = m_2 a_2 = m_2(2a_1) = 2m_2 a_1$.
For block m_1 on movable pulley: $m_1 g - 2T = m_1 a_1$.
Substituting $T = 2m_2 a_1$:

$$m_1 g - 4m_2 a_1 = m_1 a_1 \implies a_1 = \frac{m_1 g}{m_1 + 4m_2}$$

$$a_1 = \frac{4 \times 10}{4 + 4(1)} = \frac{40}{8} = 5.0 \text{ m/s}^2$$

Sol 3. Correct Option: (d)

Acceleration of individual masses:

$$a = \frac{m_1 - m_2}{m_1 + m_2} g = \frac{6 - 4}{6 + 4} (10) = 2.0 \text{ m/s}^2$$

Since m_1 accelerates downward ($-a\hat{j}$) and m_2 upward ($+a\hat{j}$):

$$a_{\text{cm}} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) a$$

$$a_{\text{cm}} = \left(\frac{2}{10} \right) (2.0) = 0.4 \text{ m/s}^2 \text{ (downward)}$$

Sol 4. Correct Option: (a)

Position constraint: $x^2 + h^2 = l^2$.
Differentiating with respect to time:

$$2x \frac{dx}{dt} = 2l \frac{dl}{dt} \implies x v_{\text{block}} = l u$$

$$v_{\text{block}} = u \frac{l}{x} = \frac{u}{\cos \theta}$$

Sol 5. Correct Option: (b)

System acceleration:

$$a = \frac{m_1 g}{m_1 + m_2 + m_3} = \frac{20}{10} = 2.0 \text{ m/s}^2$$

Tension between m_2 and m_3 pulls mass m_3 :

$$T_{23} = m_3 a = 5 \times 2.0 = 10 \text{ N}$$

Sol 6. Correct Option: (c)

Tension in the string between A and B must accelerate the combined mass below it ($m_B + m_C$) = 5 kg upward:

$$T_{AB} - (m_B + m_C)g = (m_B + m_C)a$$

$$T_{AB} = (m_B + m_C)(g + a) = 5(10 + 2) = 60 \text{ N}$$

Sol 7. Correct Option: (d)

The string exerts a horizontal tension T and a vertical downward tension T on the corner pulley. Resultant force on the axle:

$$F_{\text{pulley}} = \sqrt{T^2 + T^2} = \sqrt{2}T$$

Sol 8. Correct Option: (a)

Net driving force along the string:

$$F_{\text{net}} = m_1 g \sin 30^\circ - m_2 g = 15 - 10 = 5 \text{ N}$$

Total mass $M = 3 + 1 = 4 \text{ kg}$.

$$a = \frac{F_{\text{net}}}{M} = \frac{5}{4} = 1.25 \text{ m/s}^2$$

Sol 9. Correct Option: (b)

The virtual work principle $\sum \vec{T}_i \cdot \delta \vec{r}_i = 0$ follows directly from the constancy of the total string length ($\sum l_i = \text{const}$) in an ideal system with massless, inextensible strings and frictionless pulleys.

Sol 10. Correct Option: (c)

For a single movable pulley supporting mass m , pulling length x_0 of the free end lifts the pulley and mass m by $y = \frac{x_0}{2}$. Differentiating twice:

$$a_m = \frac{a_0}{2}$$

Sol 11. Correct Option: (d)

Let T be the tension in the rope. For the counterweight block M : $T - Mg = Ma_M$.

For the monkey of mass m : $T - mg = ma_{\text{monkey}}$.
Since monkey accelerates at $a_{\text{rel}} = 2 \text{ m/s}^2$ relative to the rope ($a_{\text{monkey}} = a_{\text{rel}} - a_M$):

$$T - mg = m(a_{\text{rel}} - a_M)$$

Since $M = m = 30$ kg, equating tensions:

$$a_M + g = (a_{\text{rel}} - a_M) + g$$

$$2a_M = 2 \implies a_M = 1.0 \text{ m/s}^2 \text{ (upward)}$$

Sol 12. Correct Option: (a)

In the center-of-mass frame of the two-block system, maximum extension occurs when relative velocity is zero:

$$x_{\text{max}} = \frac{2\mu F}{km_2} = \frac{2m_1 F}{k(m_1 + m_2)}$$

$$x_{\text{max}} = \frac{2(2)(20)}{100(2+3)} = \frac{80}{500} = 0.16 \text{ m}$$

Sol 13. Correct Option: (b)

Kinematic constraint: $a_2 = 2a_1$. Equations of motion:

$$m_1 g - 2T = m_1 a_1 \implies 40 - 2T = 4a_1$$

$$T - m_2 g = m_2(2a_1) \implies a_1 = \frac{T - 10}{2}$$

Substituting a_1 into $40 - 2T = 4a_1$:

$$40 - 2T = 2T - 20 \implies 4T = 64 \implies T = 16.0 \text{ N}$$

Sol 14. Correct Option: (c)

Using $a_1 = \frac{m_1 - 2m_2}{m_1 + 4m_2} g$:

$$a_1 = \frac{4 - 2(1)}{4 + 4(1)}(10) = \frac{2}{8}(10) = 2.0 \text{ m/s}^2$$

Sol 15. Correct Option: (d)

Equivalent mass of the hanging Atwood system:

$$m_{\text{eff}} = \frac{4m_2 m_3}{m_2 + m_3} = \frac{4(2)(3)}{2+3} = \frac{24}{5} = 4.8 \text{ kg}$$

Acceleration of block m_1 :

$$a = \frac{m_{\text{eff}} g}{m_1 + m_{\text{eff}}} = \frac{4.8 \times 10}{5 + 4.8} = \frac{48}{9.8} \approx 4.898 \text{ m/s}^2$$

Tension on m_1 :

$$T = \frac{m_1 m_{\text{eff}} g}{m_1 + m_{\text{eff}}} = \frac{240}{9.8} \approx 28.2 \text{ N}$$

Sol 16. Correct Option: (a)

Acceleration of system:

$$a = \frac{mg}{M + m} = \frac{2 \times 10}{10 + 2} = \frac{20}{12} = \frac{5}{3} \text{ m/s}^2$$

Distance in $t = 2$ s:

$$s = \frac{1}{2} a t^2 = \frac{1}{2} \left(\frac{5}{3} \right) (4) = \frac{10}{3} \approx 3.33 \text{ m}$$

Sol 17. Correct Option: (b)

$$F_{\text{net}} = 40 - 40 \sin 45^\circ = 11.72 \text{ N.}$$

Acceleration $a = \frac{11.72}{8} = 1.465 \text{ m/s}^2$. Tension:

$$T = 4(10 - 1.465) = 4(8.535) \approx 34.1 \text{ N}$$

Sol 18. Correct Option: (c)

Since $(m_1 + m_2) = (2 + 3) = 5$ kg and $m_3 = 5$ kg, the entire system is balanced and has zero acceleration ($a = 0$). Drawing the FBD of m_1 :

$$T_{12} = m_1 g = 2 \times 10 = 20 \text{ N}$$

Sol 19. Correct Option: (d)

Let T be tension in string. For hanging mass M : $T = Mg$. For ring on the vertical rod: the upward component of tension balances ring weight mg :

$$(Mg) \cos \theta = mg \implies \cos \theta = \frac{m}{M}$$

Sol 20. Correct Option: (a)

Individual acceleration:

$$a = \frac{2m - m}{2m + m} g = \frac{g}{3}$$

Acceleration of center of mass:

$$a_{\text{cm}} = \left(\frac{2m - m}{2m + m} \right) a = \left(\frac{1}{3} \right) \left(\frac{g}{3} \right) = \frac{g}{9}$$

Velocity after time t :

$$v_{\text{cm}} = a_{\text{cm}} t = \frac{gt}{9} \text{ (downward)}$$

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